INCH-POUND
MIL-DTL-24643/51E
22 August 2002
SUPERSEDING
MIL-C-24643/51D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, MULTICONDUCTOR, TYPE LSMNW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, uncoated, AWG Size No. 18, class B stranding, per ASTM B8.
Second -	Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation or ethylene propylene rubber insulation., 0.020 inch minimum average thickness. Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.
Fourth -	The required number of conductors (see table I), cabled together with a lay not greater than 24 times the pitch diameter of the layer, except that the lay of the two outer layers for size 44 shall not be greater than 18 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to form a firm, well-rounded assembly.
Fifth -	Binder tape over each pair applied helically with overlap.
Sixth -	Cross-linked polyolefin jacket. Cable jacket thickness 0.050 inch minimum average.

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TABLE I. Details.

Military part no. M24643/51	Size	Number of conductors	Overall diameter (inches) TYPE LSMNW min	Overall diameter (inches) TYPE LSMNW max	Cold bending mandrel diameter (max inch)
-01UN	7	7	0.370	0.400	3
-02UN	10	10	0.457	0.495	3
-03UN	14	14	0.494	0.535	4
-04UN	19	19	0.545	0.590	4
-05UN	24	24	0.633	0.685	5
-06UN	30	30	0.670	0.725	5
-07UN	37	37	0.726	0.785	6
-08UN	44	44	0.823	0.890	7

EXAMINATION AND TESTS:

Basic electrical: Requirements

- Conductor Resistance - ohms/1000 feet at 25°C, maximum 7.07
- Voltage withstand - volts, root mean square, minimum 3000
- Insulation resistance - Conductor to conductor, Megohms -1000 feet, minimum 1000
- Conductor continuity ..... No failure

Group A:

- Visual and dimensional ..... No failure

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (When tested at 200°C) 50
  - Cold bending, cable - (see table I) for mandrel diameter ..... No damage
  - Tear - pounds per inch thickness, minimum (ASTM D 470) 35
- Physicals (unaged)
  - Insulation
    - Tensile strength - lb/in^2, minimum 700
    - Elongation - percent, minimum 250
  - Cross-linked polyethylene
    - Tensile strength - lb/in^2, minimum 1800
    - Elongation - percent, minimum 250
  - Jacket (cable)
    - Tensile strength - lb/in^2, minimum 1300
    - Elongation - percent, minimum 160

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Group C:
Physicals - (aged)
- Insulation
  - Ethylene propylene
    - Air oven
      - Tensile strength - percent of unaged, minimum 75
      - Elongation - percent of unaged minimum 75
  - Cross-linked polyethylene
    - Air oven
      - Tensile strength - percent of unaged, minimum 80
      - Elongation - percent of unaged minimum 80
  - Jacket (cable)
    - Air oven
      - Tensile strength - percent of unaged, minimum 60
      - Elongation - percent of unaged minimum 60
    - Hot oil immersion
      - Tensile strength - percent of unaged, minimum 50
      - Elongation - percent of unaged, minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (conductor) - method 1 only, cycles, minimum 25
- Permanence of printing (jacket) - cycles minimum 125

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50

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Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket)..... No failure

UNIT ORDERING LENGTH:

Type and size	Feet (nominal)
LSMNW-7	1500
LSMNW-10	1500
All other sizes	1000

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-049)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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